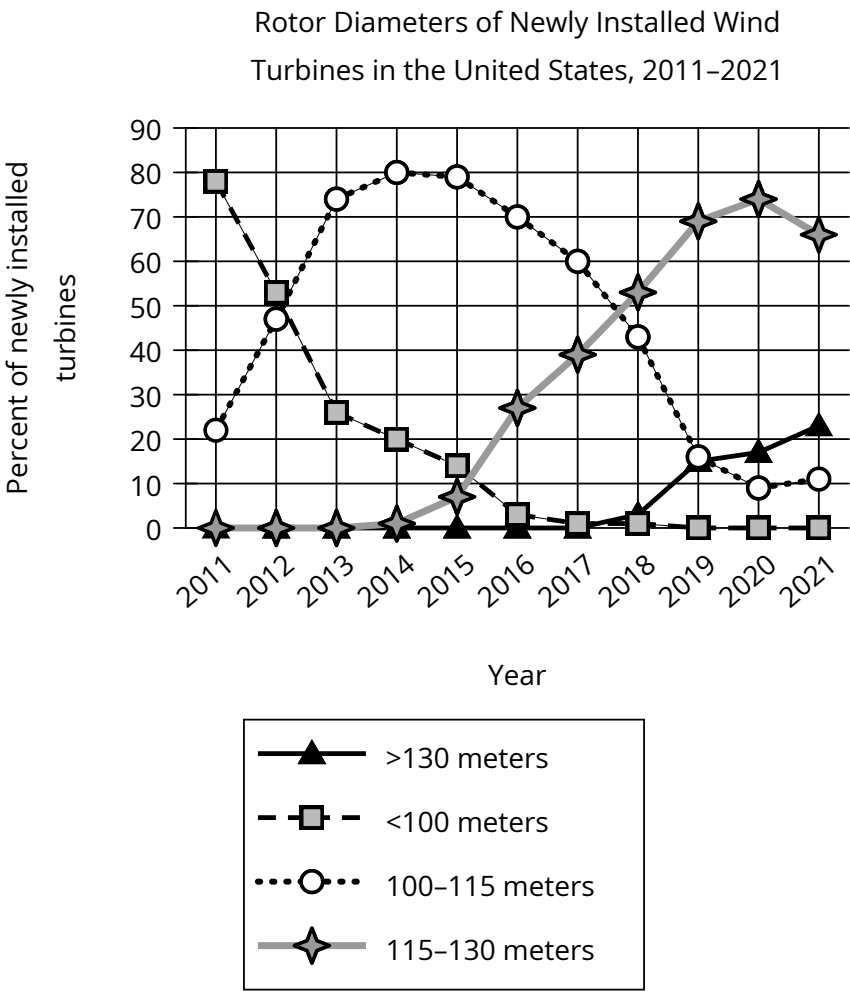


Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Medium

Question



All other things being equal, the larger a wind turbine’s rotor diameter (the diameter of the imaginary circle swept by the turbine’s rotating blades), the greater amount of energy the turbine can generate. In a research paper on wind power, a student claims that in the United States, the amount of energy generated per newly installed turbine increased substantially between 2011 and 2021.

Which choice best describes data in the graph that support the student’s claim?

- Answer
- A. The percentage of newly installed turbines with rotor diameters greater than 130 meters increased every year between 2011 and 2021.
 - B. In 2011, nearly 80% of turbines installed had rotor diameters of less than 100 meters, whereas only a little more than 20% of turbines installed that year had rotor diameters of 100–115 meters.
 - C. No turbines installed in 2011 had rotor diameters greater than 115 meters, whereas the majority of turbines installed in 2021 had rotor diameters greater than 130 meters.
 - D. Most turbines installed in 2011 had rotor diameters of less than 100 meters, whereas most turbines installed in 2021 had rotor diameters of at least 115 meters.

Correct Answer: D

Rationale

Choice D is the best answer. The text tells us that turbines with larger rotor diameters produce more energy, so if rotor diameters have generally gotten larger between 2011 and 2021, then turbines created in 2021 should produce more energy than those created in 2011.

Choice A is incorrect. This choice misreads the graph. The percentage of newly installed turbines with rotor diameters greater than 130 meters didn't show any visible increase until 2018. Choice B is incorrect. This choice doesn't justify the claim. The claim is about increasing energy output from 2011 to 2021, but this choice only discusses 2011, so it can't show evidence of change over time. Choice C is incorrect. This choice misreads the graph. In 2021, only about 25% of turbines installed in 2021 had rotor diameters greater than 130 meters.